Meta-analysis of Proportions: R Script

```
library(metafor)
library(meta)
#read in the data file
dat=read.csv("data.csv", header=T, sep=",")
#set-up file without transformation
ies=escalc(xi=cases, ni=total, data=dat, measure = "PR")
#meta-analysis with DerSimonian Laird
pes=rma(yi, vi, data=ies, method = "DL")
#view results
print(pes)
#compute and display confidence intervals of tau-squared and I-squared
confint(pes)
#create simple forest plot
pes.summary=metaprop(cases, total, authoryear, data=dat, sm="PRAW",
method.tau="DL", method.ci="NAsm")
forest(pes.summary, xlim=c(0,100),pscale=100, rightcols = FALSE,
leftcols=c("studlab", "event", "n", "effect", "ci"), leftlabs=c("Study", "Cases",
"Total", "Prevalence", "95% CI.I"), xlab="Prevalence of Non-Adherence", smlab="",
weight.study="random", squaresize=0.5, col.sqaure="black", col.squar.lines="black",
col.diamond="black", col.diamond.lines="black", pooled.totals=FALSE,
comb.fixed=FALSE, fs.hetstat=10, print.tau2=TRUE, print.Q=TRUE, print.pval.Q=TRUE,
print.I2=TRUE, digits=2)
#leave one out to identify outliers
L10=leave1out(pes); print(L10)
#visualise change in summary score with forest plot
l1o=leave1out(pes)
yi=l1o$ estimate; vi=l1o$se^2
forest(yi, vi, slab=paste(dat$author, dat$year, sep=","), refline=pes$b, xlab =
"Summary proportions leaving out each study")
#see which studies are influential using leave-one-out diagnostics
inf=influence(pes)
print(inf)
plot(inf)
```